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United States Patent Application Publication

20250287501

Kind Code

A1

Publication Date

September 11, 2025

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CIRCUIT BOARD ASSEMBLY AND PREPARATION METHOD FOR CIRCUIT BOARD ASSEMBLY

Abstract

A circuit board assembly and a preparation method for a circuit board assembly are provided. The circuit board assembly includes a printed circuit board (101) and M pads, the M pads are provided at intervals on the printed circuit board (101), first vias (103) are provided on N pads (102) in the M pads, a second via (104) is further provided at a position between any two adjacent pads (102) in the N pads (102) on the printed circuit board (101), a center of the second via (104) deviates from a connection line between centers of the any two adjacent pads (102), both M and N are integers greater than 1, and N is less than or equal to M.

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Family ID: 85338400

Appl. No.: 19/216964

Filed: May 23, 2025

Foreign Application Priority Data

CN

202211493981.9

Nov. 25, 2022

Related U.S. Application Data

parent WO continuation PCT/CN2023/132506 20231120 PENDING child US 19216964

Publication Classification

Int. Cl.: H05K1/11 (20060101); H05K3/40 (20060101)

U.S. Cl.:

CPC **H05K1/112** (20130101); **H05K3/4007** (20130101); H05K2201/0939 (20130101);
H05K2201/09409 (20130101); H05K2201/09481 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of PCT International Application No. PCT/CN2023/132506 filed on Nov. 20, 2023, which claims priority to Chinese Patent Application No. 202211493981.9, filed in China on Nov. 25, 2022, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] This application pertains to the field of chip technologies, and specifically relates to a circuit board assembly and a preparation method for a circuit board assembly.

BACKGROUND

[0003] With development of electronic technologies, consumers have increasingly high requirements on charging speeds of electronic devices such as mobile phones. Charging power of an electronic device increases from 18 W of common charging to 66 W or 80 W of fast charging and then to current 120 W or even 200 W. It can be learned that increasing the charging power is a key to improving a charging speed of the electronic device. When the charging power of the electronic device is increased, a current that needs to pass through a chip in a charging module increases accordingly from 2 A of common charging to 10 A of super charging or even a larger charging current.

[0004] When a printed circuit board (PCB) is designed, a hole is usually punched in a pad and routing is usually performed on a surface layer, to enhance an overcurrent capability. However, in a case that power is 120 W or even 200 W, a current can reach 10 A, and an overcurrent capability requirement for a large current during high-power charging is difficult to meet by punching the hole in the pad and performing routing on the surface layer. Therefore, a solution for providing vias between pads is proposed.

[0005] However, in the related technology, the solution for providing vias between pads improves an overcurrent capability for a large current during high-power charging, but an actually manufactured PCB has obvious deficiency in soldering reliability. For example, peripheries of the pads are occupied by traces and vias, resulting in a case such as a short circuit caused by pseudo soldering or solder bridging between the pads. It can be learned that reliability of a circuit board assembly is poor after an overcurrent capability for a current is improved.

SUMMARY

[0006] An objective of embodiments of this application is to provide a circuit board assembly and a preparation method for a circuit board assembly.

[0007] According to a first aspect, an embodiment of this application provides a circuit board assembly, including a printed circuit board and M pads, where the M pads are provided at intervals on the printed circuit board, first vias are provided on N pads in the M pads, a second via is further provided at a position between any two adjacent pads in the N pads on the printed circuit board, a center of the second via deviates from a connection line between centers of the any two adjacent pads, both M and N are integers greater than 1, and N is less than or equal to M.

[0008] According to a second aspect, an embodiment of this application further provides a preparation method for a circuit board assembly, where the method includes: [0009] obtaining a first to-be-prepared printed circuit board, where M pads are provided at intervals on the first to-be-prepared printed circuit board, and first vias are provided on N pads in the M pads; and [0010]

punching a second via at a position between any two adjacent pads in the N pads on the first to-be-prepared printed circuit board, to obtain a printed circuit board, where a center of the second via deviates from a connection line between centers of the any two adjacent pads, where [0011] both M and N are integers greater than 1, and N is less than or equal to M.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. 1 is a diagram 1 of a structure of a circuit board assembly according to an embodiment of this application;

[0013] FIG. 2 is a diagram 1 of a physical product of a circuit board assembly according to an embodiment of this application;

[0014] FIG. 3 is a diagram 2 of a physical product of a circuit board assembly according to an embodiment of this application;

[0015] FIG. 4 is a diagram of a connection relationship between a circuit board assembly and a chip according to an embodiment of this application;

[0016] FIG. 5 is a diagram 2 of a structure of a circuit board assembly according to an embodiment of this application;

[0018] FIG. 6 is a diagram 3 of a structure of a circuit board assembly according to an embodiment of this application;

[0019] FIG. 7 is a diagram 4 of a structure of a circuit board assembly according to an embodiment of this application; and

[0020] FIG. 8 is a flowchart of a preparation method for a circuit board assembly according to an embodiment of this application.

DESCRIPTION OF EMBODIMENTS

[0021] The following clearly describes the technical solutions in embodiments of this application with reference to the accompanying drawings in embodiments of this application. It is clear that the described embodiments are some rather than all of embodiments of this application. Based on embodiments of this application, all other embodiments obtained by a person of ordinary skill in the art fall within the protection scope of this application.

[0022] The terms “first”, “second”, and the like in the specification and claims of this application are used to distinguish between similar objects, but are not used to describe a specific sequence or order. It should be understood that data used in this way may be interchangeable in an appropriate case, so that embodiments of this application can be implemented in a sequence other than those shown or described herein, objects distinguished by “first”, “second”, and the like are generally of one type, and a quantity of objects is not limited. For example, there may be one or more first objects. In addition, in the specification and claims, “and/or” represents at least one of connected objects, and the character “/” generally represents an “or” relationship between associated objects.

[0023] Refer to FIG. 1. FIG. 1 is a diagram 1 of a structure of a circuit board assembly according to an embodiment of this application. As shown in FIG. 1, the circuit board assembly includes a printed circuit board 101 and M pads, the M pads are provided at intervals on the printed circuit board 101, first vias 103 are provided on N pads 102 in the M pads, a second via 104 is further provided at a position between any two adjacent pads 102 in the N pads 102 on the printed circuit board 101, a center of the second via 104 deviates from a connection line between centers of the any two adjacent pads 102, both M and N are integers greater than 1, and N is less than or equal to M.

[0024] The M pads on the printed circuit board 101 further include an independent pad 105, that is, no first via 103 is provided on the pad 105.

[0025] The M pads may be provided in an array on the printed circuit board 101, to improve

uniformity of spacings between the pads, and facilitate providing of the second via **104** between the pads subsequently.

[0026] Sizes of the first via **103** and the second via **104** may be the same, and copper foils may be provided on the first via **103** and the second via **104**, to enhance conductivity.

[0027] In this embodiment of this application, the M pads are provided at intervals on the printed circuit board **101**, the first vias **103** are provided on the N pads **102** in the M pads, and the second via **104** is provided at the position between the any two adjacent pads **102** in the N pads **102**, to improve an overcurrent capability for a current. In addition, the center of the second via **104** deviates from the connection line between the centers of the any two adjacent pads **102**. In this way, a distance between the second via **104** and the pad **102** is increased compared with that in a manner in which a second via is provided on a connection line between centers of any two adjacent pads, thereby reducing occupation of the pad **102** by the second via **104**, improving integrity of the pad **102**, and then improving reliability of the circuit board assembly.

[0028] Optionally, all the M pads on the printed circuit board **101** may be non-solder mask defined pads (NSMD).

[0029] As shown in FIG. 3 and FIG. 4, each pad (including the pad **102** and the pad **105**) in the M pads includes a pad body **1022** and a solder mask groove **1021**, the solder mask groove **1021** surrounds the pad body **1022**, and the solder mask groove **1021** is located between the pad body **1022** and the second via **104**, that is, a loop of solder mask groove **1021** is provided around the pad body **1022**. The solder mask groove **1021** may include a gap between ink **107** and the pad body **1022**. The pad body **1022** and the surrounding ink **107** may be separated by the solder mask groove **1021** for a specific distance. When the pad body **1022** is soldered with a material end (for example, a pad **401** on a chip), excess solder paste may flow into the solder mask groove **1021**, to reduce occurrence of solder bridging between adjacent pads.

[0030] The ink **107** may be a barrier layer that is provided on the printed circuit board and that has acid corrosion resistance and alkali corrosion resistance. The ink **107** can prevent circuit corrosion, and protect a circuit in an etching process.

[0031] The second via **104** provided on the printed circuit board **101** is located between the any two adjacent pads **102** in the N pads **102**, and is likely to occupy the adjacent pads **102**. Occupation of the solder mask groove **1021** is especially significant. Due to occupation by the second via **104**, the solder mask groove **1021** is only partially retained. When excess solder paste cannot flow into the solder mask groove **1021**, the solder paste is prone to stacking, resulting in a short circuit caused by solder bridging between different pads, thereby causing a function failure of a terminal device.

[0032] In an example, the second via **104** is provided between the any two adjacent pads **102** in the N pads **102**, and the center of the second via **104** deviates from the connection line between the centers of the any two adjacent pads **102**, to increase a distance between the second via **104** and the solder mask groove **1021**, reduce occupation of the solder mask groove **1021** by the second via **104**, and improve integrity of the pad **102** and the solder mask groove **1021**. In this way, a form of a solder joint is improved, and problems of pseudo soldering and solder bridging during soldering are resolved, thereby improving soldering reliability of the PCB.

[0033] Optionally, the first via **103** is electrically connected to the second via **104** through a trace **106**.

[0034] In an example, the first vias **103** are provided on the N pads **102** in the M pads, the second via **104** is provided at the position between the any two adjacent pads **102** in the N pads **102**, and the trace **106** is provided between the first via **103** and the second via **104**, so that signal transmission can be performed between the adjacent pads **102** through the trace **106**. The center of the second via **104** deviates from the connection line between the centers of the any two adjacent pads **102**, so that the second via **104** is staggered between the any two adjacent pads **102**. A trace **106** led from a first via **103** on each of the two adjacent pads **102** may extend toward the position

of the second via **104** for electrical connection to the second via **104**, to reduce a redundant trace between the pads **102**, and avoid a case that the trace **106** and the second via **104** occupy the pad **102** in different directions when the trace **106** and the second via **104** are respectively located in the different directions of the pad **102**, thereby improving integrity of the pad **102** and the solder mask groove **1021**. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0035] Optionally, the second via **104** is connected to one of the any two adjacent pads **102** in the N pads **102** through a first trace, the second via **104** is connected to the other of the any two adjacent pads **102** in the N pads **102** through a second trace, and an included angle between the first trace and the second trace is greater than or equal to 90 degrees.

[0036] In an example, the second via **104** is provided between the any two adjacent pads **102** in the N pads **102**, the center of the second via **104** deviates from the connection line between the centers of the any two adjacent pads **102**, and the trace **106** includes the first trace and the second trace. In this way, one of the any two adjacent pads **102** in the N pads **102** may be electrically connected to the second via **104** through extension of the first trace, and the other of the any two adjacent pads **102** in the N pads **102** may be electrically connected to the second via **104** through extension of the second trace, and the included angle between the first trace and the second trace is greater than or equal to 90 degrees, so that a case that the trace **106** and the second via **104** occupy the pad **102** in different directions is reduced, thereby improving integrity of the pad **102** and the solder mask groove **1021**. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0037] Optionally, the second via **104** is provided in a central region of any L adjacently provided pads **102** in the N pads **102**. L is an integer greater than 1 and less than or equal to N.

[0038] In an example, as shown in FIG. 5, the second via **104** may be provided in a central region of any three adjacently provided pads **102** in the N pads **102**, the three pads **102** may be distributed in a form of a triangle, and the second via **104** may be located in a central region of the triangle. In other words, the center of the second via **104** deviates from a connection line between centers of any two adjacent pads **102** in the three pads **102**, so that occupation of any one of the three pads **102** by the second via **104** is reduced. In addition, a trace **106** led from a first via **103** on each of the three pads **102** may extend toward the position of the second via **104** for electrical connection to the second via **104**, so that a case that the trace **106** and the second via **104** occupy the pad **102** in different directions is reduced, thereby improving regularity of the pad **102**, improving a form of a solder joint, and reducing a case that a stress resistance capability of the circuit board assembly is poor because an area of the pad **102** on the PCB is large, a size of a solder ball at a material end is small, and the solder joint is in an inverted funnel shape. In addition, integrity of the solder mask groove **1021** is improved, and stacking that occurs when excess solder paste cannot flow into the solder mask groove **1021** is reduced, so that reliability of the circuit board assembly is improved.

[0039] The three pads **102** may be distributed in a form of a regular triangle, and the second via **104** may be located at a position of a center point of the regular triangle, so that the center of the second via **104** has a same first distance D1 to a center of each of the three pads **102**. In this way, in a case that the second via **104** interferes with the pad **102** and/or the second via **104** interferes with the solder mask groove **1021**, effect of the same second via **104** on all of the three pads **102** can be dispersed. In addition, a trace **106** led from a first via **103** on each of the three pads **102** may extend toward the position of the second via **104** for electrical connection to the second via **104**, to reduce a redundant trace between the three pads **102**. Because each of the three pads **102** has the same first distance DI to the second via **104**, the trace **106** between each pad **102** and the second via **104** has a uniform length, so that regularity of the pad **102** and integrity of the solder mask groove **1021** are further improved. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0040] In another example, the second via **104** may be provided in a central region of any four

adjacently provided pads **102** in the N pads **102**, the four pads **102** may be distributed in a form of a rectangle, and the second via **104** may be located in a central region of the rectangle. In other words, the center of the second via **104** deviates from a connection line between centers of any two adjacent pads **102** in the four pads **102**, so that occupation of any one of the four pads **102** by the second via **104** is reduced. In addition, a trace **106** led from a first via **103** on each of the four pads **102** may extend toward the position of the second via **104** for electrical connection to the second via **104**, and pads **102** at diagonal positions may also be connected by sharing a same trace **106**, thereby reducing a redundant trace between the four pads **102**, and reducing a case that the trace **106** and the second via **104** occupy the pad **102** in different directions. In this way, regularity of the pad **102** is improved, integrity of the solder mask groove **1021** is improved, and stacking that occurs when excess solder paste cannot flow into the solder mask groove **1021** is reduced, so that reliability of the circuit board assembly is improved.

[0041] The four pads **102** may be distributed in a form of a square, and the second via **104** may be located at a position of an intersection point of diagonals of the square, so that the center of the second via **104** has a same first distance **D1** to a center of each of the four pads **102**. In this way, in a case that the second via **104** interferes with the pad **102** and/or the second via **104** interferes with the solder mask groove **1021**, effect of the same second via **104** on all of the four pads **102** can be dispersed. In addition, the trace **106** between each pad **102** and the second via **104** has a uniform length, so that regularity of the pad **102** and integrity of the solder mask groove **1021** are further improved. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0042] In another example, as shown in FIG. 6, the second via **104** may be provided in a central region of any five adjacently provided pads **102** in the N pads **102**, the five pads **102** may be distributed in a form of a pentagon, and the second via **104** may be located in a central region of the pentagon. In other words, the center of the second via **104** deviates from a connection line between centers of any two adjacent pads **102** in the five pads **102**, so that occupation of any one of the five pads **102** by the second via **104** is reduced. In addition, a trace **106** led from a first via **103** on each of the five pads **102** may extend toward the position of the second via **104** for electrical connection to the second via **104**, thereby reducing a redundant trace between the five pads **102**, and reducing a case that the trace **106** and the second via **104** occupy the pad **102** in different directions. In this way, regularity of the pad **102** is improved, integrity of the solder mask groove **1021** is improved, and stacking that occurs when excess solder paste cannot flow into the solder mask groove **1021** is reduced, so that reliability of the circuit board assembly is improved.

[0043] The five pads **102** may be distributed in a form of a regular pentagon, and the second via **104** may be located at a position of a center point of the regular pentagon, so that the center of the second via **104** has a same first distance **D1** to a center of each of the five pads **102**. In this way, in a case that the second via **104** interferes with the pad **102** and/or the second via **104** interferes with the solder mask groove **1021**, effect of the same second via **104** on all of the five pads **102** can be dispersed. In addition, the trace **106** between each pad **102** and the second via **104** has a uniform length, so that regularity of the pad **102** and integrity of the solder mask groove **1021** are further improved. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0044] In another example, as shown in FIG. 7, the second via **104** may be provided in a central region of any six adjacently provided pads **102** in the N pads **102**, the six pads **102** may be distributed in a form of a hexagon, and the second via **104** may be located in a central region of the hexagon. In other words, the center of the second via **104** deviates from a connection line between centers of any two adjacent pads **102** in the six pads **102**, so that occupation of any one of the six pads **102** by the second via **104** is reduced. In addition, a trace **106** led from a first via **103** on each of the six pads **102** may extend toward the position of the second via **104** for electrical connection to the second via **104**, and pads **102** at diagonal positions may also be connected by sharing a same

trace **106**, thereby reducing a redundant trace between the six pads **102**, and reducing a case that the trace **106** and the second via **104** occupy the pad **102** in different directions. In this way, regularity of the pad **102** is improved, integrity of the solder mask groove **1021** is improved, and stacking that occurs when excess solder paste cannot flow into the solder mask groove **1021** is reduced, so that reliability of the circuit board assembly is improved.

[0045] The six pads **102** may be distributed in a form of a regular hexagon, and the second via **104** may be located at a position of an intersection point of diagonals of the regular hexagon, so that the center of the second via **104** has a same first distance **D1** to a center of each of the six pads **102**. In this way, in a case that the second via **104** interferes with the pad **102** and/or the second via **104** interferes with the solder mask groove **1021**, effect of the same second via **104** on all of the six pads **102** can be dispersed. In addition, the trace **106** between each pad **102** and the second via **104** has a uniform length, so that regularity of the pad **102** and integrity of the solder mask groove **1021** are further improved. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0046] It should be noted that the second via **104** is provided in the central region of the any **L** adjacently provided pads **102** in the **N** pads **102**, and **L** may alternatively be another quantity, for example, 8, 9, or 12, and may be adjusted according to an actual situation. Same technical effects can be achieved, and details are not described herein again.

[0047] There is the first distance **D1** between the center of the second via **104** and a center of each of the **L** pads **102**, there is a second distance **D2** between centers of any two adjacent pads **102** in the **L** pads **102**, and a ratio of the first distance **D1** to the second distance **D2** is greater than 1:2, to increase a distance between the second via **104** and the pad **102**, reduce occupation of the pad **102** and/or the solder mask groove **1021** by the second via **104**, and improve integrity of the pad **102** and the solder mask groove **1021**. In this way, a form of a solder joint is improved, and problems of pseudo soldering and solder bridging during soldering are resolved, thereby improving reliability of the circuit board assembly.

[0048] Optionally, a size of the trace **106** is less than or equal to a size of the pad **102**, and/or the size of the trace **106** is less than or equal to a size of the second via **104**.

[0049] In embodiments of this application, a width of the trace **106** between the pads **102** in the **M** pads is reduced, so that the size of the trace **106** is less than or equal to the size of the pad **102**, and/or the size of the trace **106** is less than or equal to the size of the second via **104**, thereby reducing a case that large-area interference between the trace **106** and the pad **102** leads to poor integrity of the solder mask groove **1021**. In this way, routing on a surface layer of the PCB is implemented and the via is added, so that an overcurrent capability is improved. In addition, a size and integrity of the solder mask groove **1021** are retained to a maximum extent, so that problems such as solder bridging and pseudo soldering generated during soldering are resolved.

[0050] Optionally, a quantity of second vias **104** adjacent to each of the **N** pads **102** is less than or equal to two.

[0051] In embodiments of this application, the **N** pads **102** may include a plurality of pad networks, and one pad network may include at least two pads **102**. A second via **104** may be provided between any two adjacent pads **102** in the at least two pads **102**, so that a center of the second via **104** deviates from a connection line between centers of the any two adjacent pads **102**. In addition, a trace **106** led from each of the two adjacent pads **102** extends toward a position of the second via **104** for electrical connection to the second via **104**, a width of the trace **106** may be less than or equal to a width of the pad **102**, and a quantity of second vias **104** electrically connected to each pad **102** is less than or equal to two. In this way, routing on a surface layer of the PCB is implemented and the via is added, so that an overcurrent capability is improved. In addition, a size and integrity of the solder mask groove **1021** are retained to a maximum extent, so that problems such as solder bridging and pseudo soldering generated during soldering are resolved, and reliability of the circuit board assembly is improved.

[0052] When the PCB is designed, one or more pads **102** may be connected to an independent pad **105** through a trace **106** according to an actual requirement, and a second via **104** may be provided on the trace **106** between the pad **102** and the independent pad **105**, so that a trace led from the independent pad **105** faces the second via **104**, and a trace led from the pad **102** faces the second via **104**, thereby reducing a redundant trace between the pad **102** and the independent pad **105**, and reducing occupation of the pad by the trace **106** and the second via **104**. In this way, integrity of the pad is improved. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0053] Refer to FIG. **8**. FIG. **8** is a flowchart of a preparation method for a circuit board assembly according to an embodiment of this application. The method is used to prepare the foregoing circuit board assembly. The method includes the following steps.

[0054] Step **801**: Obtain a first to-be-prepared printed circuit board, where M pads are provided at intervals on the first to-be-prepared printed circuit board, and first vias are provided on N pads in the M pads.

[0055] In this step, the first to-be-prepared printed circuit board may be a to-be-processed PCB that has not been provided with a second via. The M pads are provided on the first to-be-prepared printed circuit board, the M pads may be provided in an array on the first to-be-prepared printed circuit board, and the first vias are provided on the N pads in the M pads, to facilitate subsequent soldering with a material end by using the pads.

[0056] Step **802**: Punch a second via at a position between any two adjacent pads in the N pads on the first to-be-prepared printed circuit board, to obtain a printed circuit board, where a center of the second via deviates from a connection line between centers of the any two adjacent pads.

[0057] Both M and N are integers greater than 1, and N is less than or equal to M.

[0058] In this step, an overcurrent capability of the circuit board assembly may be improved by providing the via, the second via is located between the any two adjacent pads, and the center of the second via deviates from the connection line between the centers of the any two adjacent pads, to reduce occupation of the pad by the second via, and improve a form of a solder joint.

[0059] In this embodiment of this application, in a PCB design process, the M pads are provided on the first to-be-prepared printed circuit board, the first vias are provided on the N pads in the M pads, and the second via is punched at the position between the any two adjacent pads in the N pads on the first to-be-prepared printed circuit board, to improve an overcurrent capability for a current. In addition, the center of the second via deviates from the connection line between the centers of the any two adjacent pads, to obtain the printed circuit board. In the printed circuit board, a distance between the second via and the pad is increased, thereby reducing occupation of the pad by the second via, improving integrity of the pad, and then improving reliability of the circuit board assembly.

[0060] The punching a second via at a position between any two adjacent pads in the N pads on the first to-be-prepared printed circuit board, to obtain a printed circuit board, where a center of the second via deviates from a connection line between centers of the any two adjacent pads includes:

[0061] punching the second via at the position between the any two adjacent pads in the N pads on the first to-be-prepared printed circuit board, to obtain a second to-be-prepared printed circuit board, where the center of the second via deviates from the connection line between the centers of the any two adjacent pads; and [0062] performing line connection on the second to-be-prepared printed circuit board, to obtain the printed circuit board, where a trace between any two adjacent pads on the printed circuit board extends toward the position of the second via for electrical connection to the second via.

[0063] In this way, a trace may be provided between two adjacent pads, and a trace led from each of the two adjacent pads extends toward the position of the second via for electrical connection to the via, to reduce a redundant trace between the pads, and avoid a case that the trace and the via occupy the pad in different directions when the trace and the via are respectively located in the

different directions of the pad, thereby improving integrity of the pad. The circuit board assembly has high reliability while an overcurrent capability for a current is improved.

[0064] Optionally, the punching a second via at a position between any two adjacent pads in the N pads on the first to-be-prepared printed circuit board, to obtain a printed circuit board, where a center of the second via deviates from a connection line between centers of the any two adjacent pads includes: [0065] in a case that a target region on the first to-be-prepared printed circuit board meets a via punching condition, punching the second via in the target region, to obtain the printed circuit board.

[0066] The target region is the position between the any two adjacent pads in the N pads, and the target region deviates from the connection line between the centers of the any two adjacent pads.

[0067] In this embodiment of this application, when the second via is punched in the target region on the first to-be-prepared printed circuit board, interference of the second via to adjacent pads in the N pads is considered. In a case that the interference of the second via to the pads is less than a target threshold, that is, a spacing between any two adjacent pads in the N pads is large, the second via may be provided between the two pads. On the contrary, no via is punched. In this way, integrity of the pad is improved.

[0068] It should be noted that, in this specification, the term “include”, “have”, or any other variant thereof is intended to cover non-exclusive inclusion, so that a process, method, article, or apparatus that includes a series of elements includes not only those elements but also other elements that are not explicitly listed, or includes elements inherent to such a process, method, article, or apparatus. In a case without more restrictions, for an element limited by the statement “include a . . .”, a process, method, article, or apparatus that includes the element may further include another same element. In addition, it should be noted that the scope of the method and apparatus in embodiments of this application is not limited to performing a function in an order shown or discussed, and may further include performing a function in a basically simultaneous manner or in a reverse order according to a related function. For example, the described method may be performed in a different order, and various steps may be added, omitted, or combined. In addition, features described with reference to some examples may be combined in other examples.

[0069] Embodiments of this application are described above with reference to the accompanying drawings. However, this application is not limited to the foregoing specific implementations. The foregoing specific implementations are merely an example but not a limitation. Many forms that can be made by a person of ordinary skill in the art without departing from the principle of this application and the protection scope of the claims fall within the protection scope of this application.

Claims

1. A circuit board assembly, comprising a printed circuit board and M pads, wherein the M pads are provided at intervals on the printed circuit board, first vias are provided on N pads in the M pads, a second via is further provided at a position between any two adjacent pads in the N pads on the printed circuit board, a center of the second via deviates from a connection line between centers of the any two adjacent pads, both M and N are integers greater than 1, and N is less than or equal to M.

2. The circuit board assembly according to claim 1, wherein the second via is provided in a central region of any L adjacently provided pads in the N pads, and L is an integer greater than 1 and less than or equal to N.

3. The circuit board assembly according to claim 2, wherein there is a first distance between the center of the second via and a center of each of the L pads, there is a second distance between centers of any two adjacent pads in the L pads, and a ratio of the first distance to the second distance is greater than 1:2.

4. The circuit board assembly according to claim 1, wherein the first via and the second via are electrically connected through a trace.
 5. The circuit board assembly according to claim 2, wherein the first via and the second via are electrically connected through a trace.
 6. The circuit board assembly according to claim 3, wherein the first via and the second via are electrically connected through a trace.
 7. The circuit board assembly according to claim 4, wherein a size of the trace is less than or equal to a size of the pad, and/or the size of the trace is less than or equal to a size of the second via.
 8. The circuit board assembly according to claim 1, wherein the second via is connected to one of the any two adjacent pads in the N pads through a first trace, the second via is connected to the other of the any two adjacent pads in the N pads through a second trace, and an included angle between the first trace and the second trace is greater than or equal to 90 degrees.
 9. The circuit board assembly according to claim 1, wherein each of the M pads is a non-solder mask defined pad.
 10. The circuit board assembly according to claim 9, wherein each of the M pads comprises a pad body and a solder mask groove, the solder mask groove surrounds the pad body, and the solder mask groove is located between the pad body and the second via.
 11. The circuit board assembly according to claim 10, wherein ink is further provided on the printed circuit board, and a gap between the ink and the pad body forms the solder mask groove.
 12. A preparation method for a circuit board assembly, wherein the method comprises: obtaining a first to-be-prepared printed circuit board, wherein M pads are provided at intervals on the first to-be-prepared printed circuit board, and first vias are provided on N pads in the M pads; and punching a second via at a position between any two adjacent pads in the N pads on the first to-be-prepared printed circuit board, to obtain a printed circuit board, wherein a center of the second via deviates from a connection line between centers of the any two adjacent pads, wherein both M and N are integers greater than 1, and N is less than or equal to M.
 13. The method according to claim 12, wherein the second via is provided in a central region of any L adjacently provided pads in the N pads, and L is an integer greater than 1 and less than or equal to N.
 14. The method according to claim 13, wherein there is a first distance between the center of the second via and a center of each of the L pads, there is a second distance between centers of any two adjacent pads in the L pads, and a ratio of the first distance to the second distance is greater than 1:2.
 15. The method according to claim 12, wherein the first via and the second via are electrically connected through a trace.
 16. The method according to claim 15, wherein a size of the trace is less than or equal to a size of the pad, and/or the size of the trace is less than or equal to a size of the second via.
 17. The method according to claim 12, wherein the second via is connected to one of the any two adjacent pads in the N pads through a first trace, the second via is connected to the other of the any two adjacent pads in the N pads through a second trace, and an included angle between the first trace and the second trace is greater than or equal to 90 degrees.
 18. The method according to claim 12, wherein each of the M pads is a non-solder mask defined pad.
 19. The method according to claim 18, wherein each of the M pads comprises a pad body and a solder mask groove, the solder mask groove surrounds the pad body, and the solder mask groove is located between the pad body and the second via.
 20. The method according to claim 19, wherein ink is further provided on the printed circuit board, and a gap between the ink and the pad body forms the solder mask groove.
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